Integrating Machine-Learning $PM_{2.5}$ Forecasting, AQI Classification and Epidemiological Health-Burden Estimation in Ghana

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Abstract

Air pollution poses a major public health challenge globally, with fine particulate matter ($PM_{2.5}$) contributing substantially to premature mortality and disease burden. Despite increasing concern over air quality in developing countries, integrated frameworks that combine pollutant forecasting, air quality risk communication, and health burden assessment remain limited in Sub-Saharan Africa. This study developed a machine learning-based framework for forecasting $PM_{2.5}$ concentrations, classifying Air Quality Index (AQI) levels, and estimating associated health impacts using data collected from the Afri-SET air quality monitoring station at the University of Ghana between January 2022 and December 2023. Multiple machine learning and deep learning models were evaluated for hourly $PM_{2.5}$ prediction and AQI classification. Results showed that the Random Forest model achieved the best forecasting performance with an R^2 of 0.8041, RMSE of 3.1339 $\mu g/m^3$, and MAE of 2.0058 $\mu g/m^3$, outperforming deep learning models. For AQI classification, Support Vector Machine (SVM) was used together with class-weighted and SMOTE to handle class imbalance where class-weighted achieved strong stability compared to SMOTE, with strong predictive capability across multiple air quality categories. Analysis of observed pollution levels revealed frequent exceedances of World Health Organization $PM_{2.5}$ guidelines, particularly during the Harmattan season. Using exposure-response relationships from epidemiological literature, the study estimated approximately 11,230 attributable deaths and 370,339 disability-adjusted life years associated with long-term $PM_{2.5}$ exposure. The findings demonstrate the potential of machine learning-driven environmental health frameworks to support air quality management, public health planning, and early warning systems in resource-constrained settings.

Keywords: $PM_{2.5}$; air-quality forecasting; machine learning; Air Quality Index; environmental epidemiology; Ghana.

1. Introduction

Ambient air pollution remains a major environmental and public-health challenge worldwide, with fine particulate matter ($PM_{2.5}$) receiving particular attention because of its ability to penetrate deeply into the respiratory system and its strong association with cardiovascular and respiratory diseases (WHO, 2021; HEI, 2020). Rapid urbanisation, increasing transport emissions, biomass burning, dust resuspension, limited monitoring infrastructure, and seasonal meteorological variability further exacerbate air-quality challenges across many African cities (Amegah & Agyei-Mensah, 2017).

In Ghana, air-quality monitoring remains geographically limited relative to the spatial and temporal variability of urban pollution. Evidence from Accra indicates that particulate-matter concentrations vary considerably across locations and time periods due to differences in traffic density, land-use characteristics, local emission sources, and meteorological conditions (Alli et al., 2021). These complexities highlight the need for reliable forecasting systems capable of supporting air-quality management and public-health decision making.

Machine-learning approaches have emerged as valuable tools for air-quality prediction because they can model nonlinear relationships between pollutant concentrations, meteorological variables, and temporal factors. Tree-based algorithms such as Random Forest and XGBoost effectively capture complex interactions, while deep-learning architectures including Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks can learn hierarchical patterns and temporal dependencies from sequential data (Rahman et al., 2024). However, model performance remains highly dependent on local environmental conditions, feature representation, forecast horizon, and validation strategy.

Despite advances in pollutant forecasting, AQI prediction, and environmental-health assessment, these areas are often investigated independently. Most studies focus primarily on forecasting accuracy, while AQI classification and health-burden estimation are conducted separately using observed pollution measurements. Consequently, relatively few studies have developed integrated frameworks that connect pollutant forecasting, air-quality risk communication, and epidemiological assessment, particularly in data-constrained environments such as Sub-Saharan Africa (Kocak, 2025).

To address this gap, this study developed an integrated environmental-health framework using high-resolution air-quality observations from Ghana. The framework combines continuous $PM_{2.5}$ forecasting for exposure estimation, NowCast-based Air Quality Index (AQI) classification for risk communication, and epidemiological burden estimation using disease-specific concentration-response functions together with Ghana-specific mortality and disability-adjusted life year (DALY) estimates. Specifically, the study: (i) characterized the temporal behaviour of $PM_{2.5}$ concentrations; (ii) developed and evaluated machine-learning, deep-learning, and hybrid models for hourly $PM_{2.5}$ forecasting; (iii) assessed the performance of a NowCast-based Support Vector Machine (SVM) AQI classification framework under class imbalance; and (iv) estimated the mortality and DALY burden attributable to long-term $PM_{2.5}$ exposure. By integrating forecasting, classification, and health-impact assessment within a unified analytical framework, the study provides a practical approach for supporting air-quality management and public-health planning in resource-constrained settings.

2. Related Work

Air-quality dynamics are governed by complex interactions among pollutant emissions, meteorological conditions, atmospheric transport, chemical transformation, and temporal variability, producing nonlinear and nonstationary patterns that are often difficult to model using conventional statistical approaches (Alli et al., 2021). Consequently, machine learning (ML) methods have gained significant attention because they can capture nonlinear relationships and

accommodate multiple predictors without requiring strict parametric assumptions. Algorithms such as Random Forest (RF), Support Vector Machine (SVM), XGBoost, and LightGBM have demonstrated strong performance in air-quality prediction. Previous studies have reported effective performance of XGBoost in AQI prediction, while ensemble methods such as Gradient Boosting and Extra Trees Regression have been applied to particulate-matter and gaseous-pollutant modelling (Tırınk, 2025; Koçak, 2025). The predictive performance of these models can be enhanced by incorporating meteorological variables including temperature, relative humidity, and wind speed, which influence pollutant dispersion and atmospheric processes. However, conventional ML approaches do not inherently account for temporal dependencies and therefore rely heavily on engineered features such as lagged observations and rolling statistics.

Deep learning (DL) methods offer an alternative by automatically learning nonlinear representations from sequential data. Architectures such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks are particularly suitable for air-quality forecasting because they can capture temporal dependencies in pollutant concentrations (Kırelli, 2025). Hybrid approaches combining convolutional and recurrent structures have further improved predictive performance. For instance, CNN-Bi-GRU models integrate local feature extraction with temporal learning to represent complex air-quality patterns (Ghose et al., 2021). Additionally, attention mechanisms have been used to enhance deep-learning performance by emphasizing informative predictors and temporal dependencies (Ogusanwo et al., 2025). Despite these advantages, DL models typically require larger datasets, greater computational resources, and extensive hyperparameter tuning, while their limited interpretability may constrain their adoption in operational decision-support systems.

Beyond forecasting, air-pollution research increasingly focuses on understanding public-health impacts. Machine-learning approaches have been applied to estimate exposure patterns and mortality risks associated with air pollution (Singh et al., 2023), while epidemiological tools such as AirQ+ have also been used to estimate mortality attributable to $PM_{2.5}$ exposure (Koçak, 2025). Classification-based ML approaches have demonstrated potential for health-risk prediction (Alkakuri et al., 2025). Similarly, health-oriented frameworks such as the Health Risk-Based Air Quality Index (HAQI) provide integrated assessments of pollutant-related risks (Kumar et al., 2025). These developments highlight growing efforts to connect environmental monitoring with public-health outcomes.

Despite these advances, existing studies largely treat pollutant forecasting, AQI classification, and epidemiological assessment as separate research domains. Relatively few studies integrate continuous $PM_{2.5}$ forecasting with AQI classification and health-burden estimation within a single framework. Furthermore, direct exposure-outcome modelling is often constrained by limited availability of temporally aligned and location-specific health data, particularly in developing countries. Model performance also remains geographically and temporally context-dependent, while the trade-off between predictive performance, computational complexity, and interpretability continues to influence model selection. Therefore, there remains a need for integrated environmental-health frameworks that connect $PM_{2.5}$ forecasting, AQI classification, and health-burden estimation to support evidence-based air-quality management and public-health decision-making.

3. Materials and Methods

3.1 Study Area and Data Sources

This study utilized air-quality observations from the Afri-SET monitoring station located at the University of Ghana Physics Department in the Greater Accra Metropolitan Area (GAMA), Ghana. The monitoring record was treated as a site-specific high-resolution time series representing local atmospheric conditions rather than a direct representation of air quality across the entire metropolitan area.

The environmental dataset comprised 17,544 hourly observations collected between 1 January 2024 and 31 December 2025. Variables included $PM_{2.5}$, PM_{10}, temperature, relative humidity, wind speed, wind direction, and timestamp information. The health component of the study employed Ghana-specific baseline mortality and disability-adjusted life year (DALY) estimates for ischemic heart disease (IHD), stroke, lower respiratory infection (LRI), and chronic obstructive pulmonary disease (COPD) obtained from the IHME Global Burden of Disease (GBD) 2023 database.

3.2 Data Preprocessing

Missing observations occurred primarily due to monitoring interruptions. Gaps of one to three consecutive hours were imputed using time-based linear interpolation, whereas longer gaps were estimated using Random Forest regression based on available environmental and temporal predictors. Data imputation was completed prior to feature engineering and target-variable construction.

For a missing observation bounded by valid observations at times t_a and t_b, linear interpolation was calculated as:

$$y_t = y_a + \frac{t - t_a}{t_b + t_a}(y_b - y_a)$$

Random Forest imputation was estimated as:

$$y = \frac{1}{B}\sum_{b=1}^{B} T_b\,(X)$$

3.3 Feature Engineering and Selection

The forecasting target was defined as the next-hour $PM_{2.5}$ concentration:

Target_$PM_{2.5}$,t = $PM_{2.5}$,t+1.

Predictor variables included meteorological observations, cyclical temporal encodings, $PM_{2.5}$ lag variables (1, 3, 6, 12, and 24 h), rolling means (6, 12, and 24 h), a 24-h rolling standard deviation,

and interaction terms involving temperature, relative humidity, and particulate-matter concentrations.

The final modelling dataset contained 17,519 observations and 28 predictors. A reduced dataset containing 15 predictors was generated using Recursive Feature Elimination (RFE).

Feature selection involved Pearson correlation analysis, Variance Inflation Factor (VIF) diagnostics, Random Forest feature importance, and RFE. To prevent persistence effects from dominating the importance analysis, current-hour $PM_{2.5}$ was excluded during diagnostic feature evaluation. Predictor standardization was applied to neural-network models, whereas tree-based algorithms were trained using unscaled inputs.

3.4 Model Development and Validation

A dual-pipeline framework was implemented consisting of a continuous $PM_{2.5}$ forecasting module and an Air Quality Index (AQI) classification module. The forecasting pipeline was designed to estimate pollutant exposure for subsequent epidemiological assessment, whereas the classification pipeline was used to communicate short-term air-quality conditions.

For $PM_{2.5}$ forecasting, the dataset was divided using a chronological 80:20 train-test split. Model development employed five-fold expanding-window cross-validation using TimeSeriesSplit to preserve temporal ordering and prevent information leakage from future observations.

Five forecasting models were evaluated: Random Forest (RF), Extreme Gradient Boosting (XGBoost), one-dimensional Convolutional Neural Network (1D-CNN), Long Short-Term Memory (LSTM), and a hybrid XGBoost-LSTM architecture. Tree-based models were selected because of their ability to model nonlinear environmental relationships, while deep-learning architectures were included to capture temporal dependencies in sequential observations. The hybrid model was developed to assess whether combining ensemble learning with recurrent sequence modelling could improve one-hour-ahead $PM_{2.5}$ prediction.

Model performance was evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2):

Root Mean Squared Error (RMSE): Measures the magnitude of prediction errors and penalizes larger variances:

$$\boldsymbol{RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(C_i - C_{ti})\text{^}2}}$$

Mean Absolute Error (MAE): Quantifies the average absolute prediction error magnitude across the timeline:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|C_i - C_{ti}|$$

Coefficient of Determination (R^2): Quantifies the proportion of environmental variance explained by the model features:

$$R^2 = 1 - \left(\frac{(\sum_{i=1}^{n}(C_i - C_{ti})^2}{(\sum_{i=1}^{n}(C_i - C_t)^2}\right)$$

To assess the implications of forecasting uncertainty on health-impact estimates, epidemiological prediction-error propagation was also evaluated by examining the influence of exposure prediction errors on relative risk (RR), population attributable fraction (PAF), attributable mortality, and DALYs.

AQI Classification

AQI classification was performed using a NowCast representation of recent $PM_{2.5}$ conditions and a radial basis function (RBF) Support Vector Machine (SVM). Six AQI categories were considered: Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy, Very Unhealthy, and Hazardous.

Because the Moderate category constituted approximately 77.6% of observations, classification performance was assessed using Precision, Recall, Macro Recall, Macro F1-score, and confusion-matrix analysis. Class-weighted SVM and controlled Synthetic Minority Over-sampling Technique (SMOTE) were additionally evaluated to address class imbalance.

Table 3. 1. AQI Categories and $PM_{2.5}$ Breakpoints

AQI Range	Category	$PM_{2.5}$ Breakpoints	Health Implication
0 -50	Good	0.0 – 9.0	Minimal risk
51-100	Moderate	9.1 – 35.4	Acceptable; some risk for sensitive groups
101-150	Unhealthy for Sensitive Groups	35.5 – 55.4	Increased risk for vulnerable populations
151-200	Unhealthy	55.5 – 125.4	General public may experience health effects
201-300	Very Unhealthy	125.5 – 225.4	Serious health warnings
>300	Hazardous	225.5 – 325.4	Emergency conditions

Classification metrics were calculated as:

For class k, Precision_k, evaluates the model's capacity to avoid false positive alarms:

$$P_k = \frac{True\ Postive\ (TP_k)}{True\ Positives\ (TP_k) + False\ Postives\ (FP_k)}$$

Recall_k , measures the model's capacity to successfully identify hazardous events:

$$R_k = \frac{True\ Postive\ (TP_k)}{True\ Positives\ (TP_k) + False\ Negative\ (FN_k)}$$

Macro F1-Score: The macro F1-score was used to provide an overall assessment of classification performance across the AQI categories while giving equal weight to each class, irrespective of its frequency. For each class kk, the F1-score was calculated as:

$$F1_k = \frac{2(Precsion_k * Recall_k)}{Precision_k + Recall_k}$$

The macro F1-score was subsequently calculated as:

$$Macro\ F1 = \frac{1}{K}\sum_{k=1}^{k} F1_k$$

3.5 Epidemiological Health-Burden Estimation

The best-performing forecasting model was used to estimate annual $PM_{2.5}$ exposure from 8,760 hourly predictions. Disease-specific relative risks were calculated using log-linear concentration-response functions:

$$RR = e^{\beta(C-C_0)}$$

Where C is the annual $PM_{2.5}$ concentration, β is the disease-specific concentration-response coefficient, and C_0 is the theoretical minimum risk exposure level (TMREL). A counterfactual concentration of 5.0 μg m^{-3} was adopted (WHO, 2021).

The concentration-response coefficients were derived from peer-reviewed epidemiological studies for stroke, ischemic heart disease, lower respiratory infection, and chronic obstructive pulmonary disease.

Table 3. 2. Relative-Risk Parameters Used in Health-Burden Estimation

Disease	RR (per 10μg/m3 increase)	Derived Pipeline Constant (β)
Stroke (Cerebrovascular Disease)	1.11	0.01044

Ischemic Heart Disease (IHD)	1.15	0.01398
Lower Respiratory Infections (LRI)	1.15	0.01398
Chronic Obstructive Pulmonary Disease (COPD)	1.08	0.0077

The population attributable fraction (PAF) was calculated as:

$$PAF = \frac{RR - 1}{RR}$$

Attributable mortality and DALYs were estimated as:

DeathsAttributable; Deaths = PAF×Baseline Deaths

DALYsAttributable; DALYs = PAF×Baseline DALYs

The resulting estimates represent model-based population-level health burdens under the stated exposure-response assumptions. Because the analysis was based on observations from a single monitoring station, findings should be interpreted as a site-specific environmental-health assessment rather than a nationally representative exposure analysis.

4. Results and Discussions

4.1 Temporal Characteristics of $PM_{2.5}$ Concentrations

The 17,544-hour monitoring record revealed substantial temporal and seasonal variability in $PM_{2.5}$ concentrations. Mean concentrations were highest during the morning period (06:00-08:00), averaging 29.97 µg/m³, with a peak hourly mean of 30.39 µg/m³ observed at approximately 07:00. Concentrations during the late afternoon and early evening period (16:00-19:00) averaged 22.45 µg/m³. Mean $PM_{2.5}$ concentrations were similar between weekdays (24.40 µg/m³) and weekends (25.53 µg/m³), with no statistically significant difference detected (Mann-Whitney, $p = 0.297$).

Among the 731 daily observations, 103 days (14.09%) exceeded Ghana's 24-hour $PM_{2.5}$ standard of 35 µg/m³. The highest daily mean concentration was 227.82 µg/m³, recorded on 7 February 2024. Pollution levels were substantially higher during the November-February period, which averaged 43.60 µg/m³, compared with 15.44 µg/m³ during March-October. Exceedances of the national standard occurred on 41.49% of days during November-February but only 0.61% of days during March-October. The seasonal difference was statistically significant (Kruskal-Wallis $H = 498.43$, $p < 0.001$).

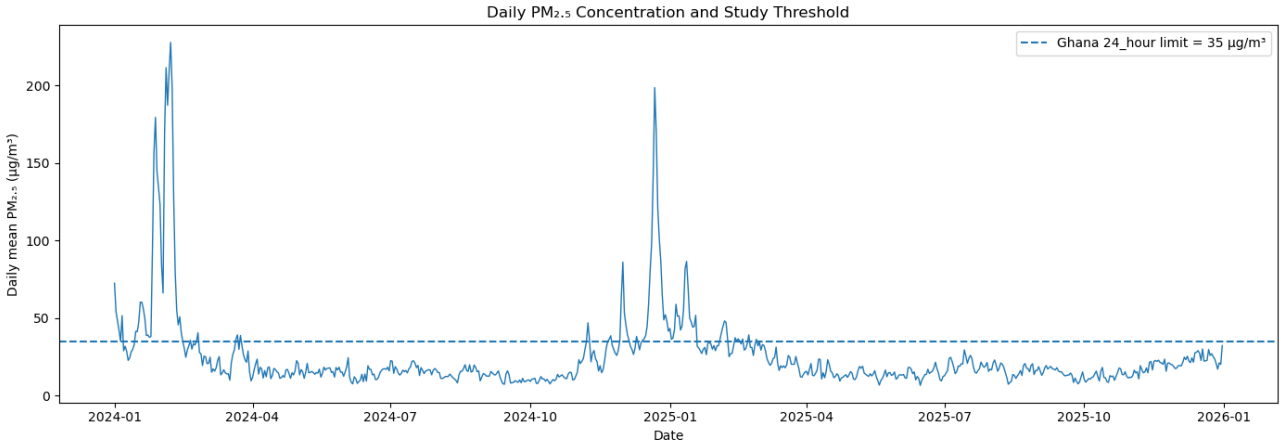


Fig.4. 1. Daily $PM_{2.5}$ concentration and the 35 µg/m³ study threshold during January 2024–December 2025.

Figure 1 illustrates daily $PM_{2.5}$ concentrations over the study period. Exceedances were concentrated primarily during the dry-season months, with several episodic pollution events exceeding 200 µg/m³, indicating substantial short-term variability that is not captured by annual averages alone.

4.2 Feature Importance and Dimensionality Reduction

Following exclusion of current-hour $PM_{2.5}$ from the diagnostic feature ranking, PM_{10} emerged as the most influential predictor, accounting for 61.20% of Random Forest importance, followed by the $PM_{2.5}$-relative humidity interaction term (18.36%). Recursive Feature Elimination (RFE) reduced the predictor space from 28 to 15 variables while retaining key particulate, meteorological, temporal, and interaction features.

As shown in **Figure 2**, both Random Forest and XGBoost identified particulate-matter variables as the dominant predictors of one-hour-ahead $PM_{2.5}$ concentrations. Lagged $PM_{2.5}$ variables, rolling statistical summaries, and meteorological predictors also contributed substantially to model performance, highlighting the importance of both pollution persistence and atmospheric conditions in short-term forecasting.

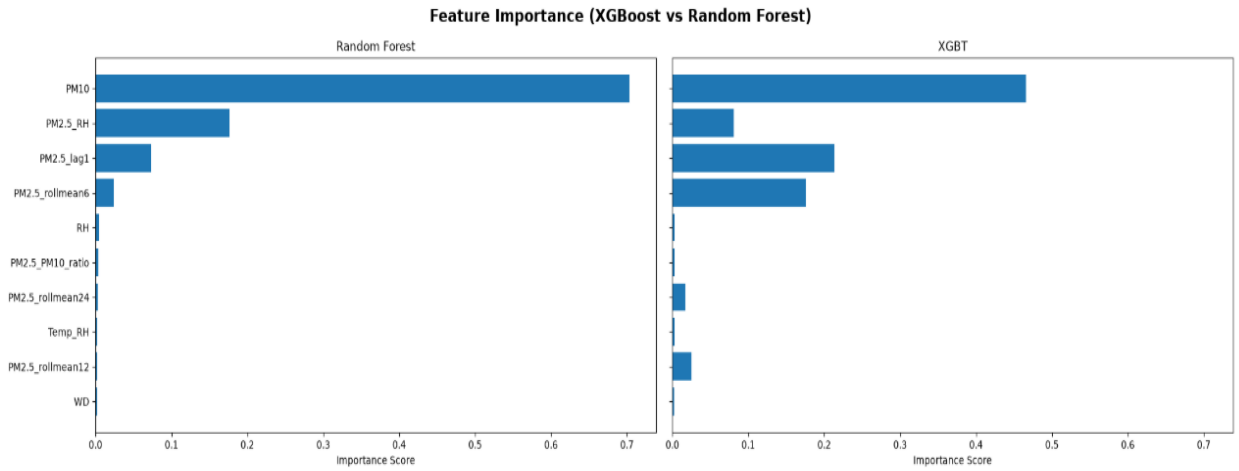


Fig.4. 2. Feature importance in one-hour-ahead $PM_{2.5}$ prediction.

4.3 $PM_{2.5}$ Forecasting Performance

Table 1 summarizes forecasting performance for the reduced feature set (15 predictors). Tree-based models achieved the strongest overall performance, with XGBoost marginally outperforming Random Forest. Deep-learning models produced substantially higher prediction errors, although the hybrid XGBoost-LSTM model achieved better performance than the standalone CNN and LSTM architectures.

Table 4. 1. Performance of $PM_{2.5}$ forecasting models using the reduced 15-predictor feature set.

Model	MAE(ug/m^3)	RMSE(ug/m^3)	R^2
Random Forest (RF)	2.0947	3.3765	0.7726
XGBoost (XGBT)	2.1662	3.3312	0.7786
CNN	3.8156	5.3592	0.4298
LSTM	3.6968	5.3050	0.4413
Hybrid XGBoost-LSTM	3.7146	5.2842	0.4457

Performance improved across all models when the full feature set (28 predictors) was used (Table 2). The optimized Random Forest achieved the best overall performance, producing the lowest prediction errors and highest explanatory power (R^2 = 0.8041). XGBoost ranked second, while the hybrid XGBoost-LSTM remained the strongest deep-learning configuration.

Table 4. 2. Performance of $PM_{2.5}$ forecasting models using the full 28-predictor feature set.

Model	MAE(ug/m^3)	RMSE(ug/m^3)	R^2
Random Forest (RF)	2.0058	3.1339	0.8041
XGBoost (XGBT)	2.0873	3.2221	0.7929

CNN	4.0068	5.5809	0.3817
LSTM	3.8719	5.6419	0.3681
Hybrid XGBoost-LSTM	3.6164	5.1861	0.4661

Overall, the inclusion of additional predictors improved forecasting accuracy, indicating the value of particulate, temporal, meteorological, and interaction features for short-term $PM_{2.5}$ prediction.

4.4 AQI Classification Performance

The Moderate AQI category accounted for 77.59% of observations, whereas the Very Unhealthy and Hazardous categories represented only 1.63% and 0.25%, respectively, indicating substantial class imbalance (Figure 3).

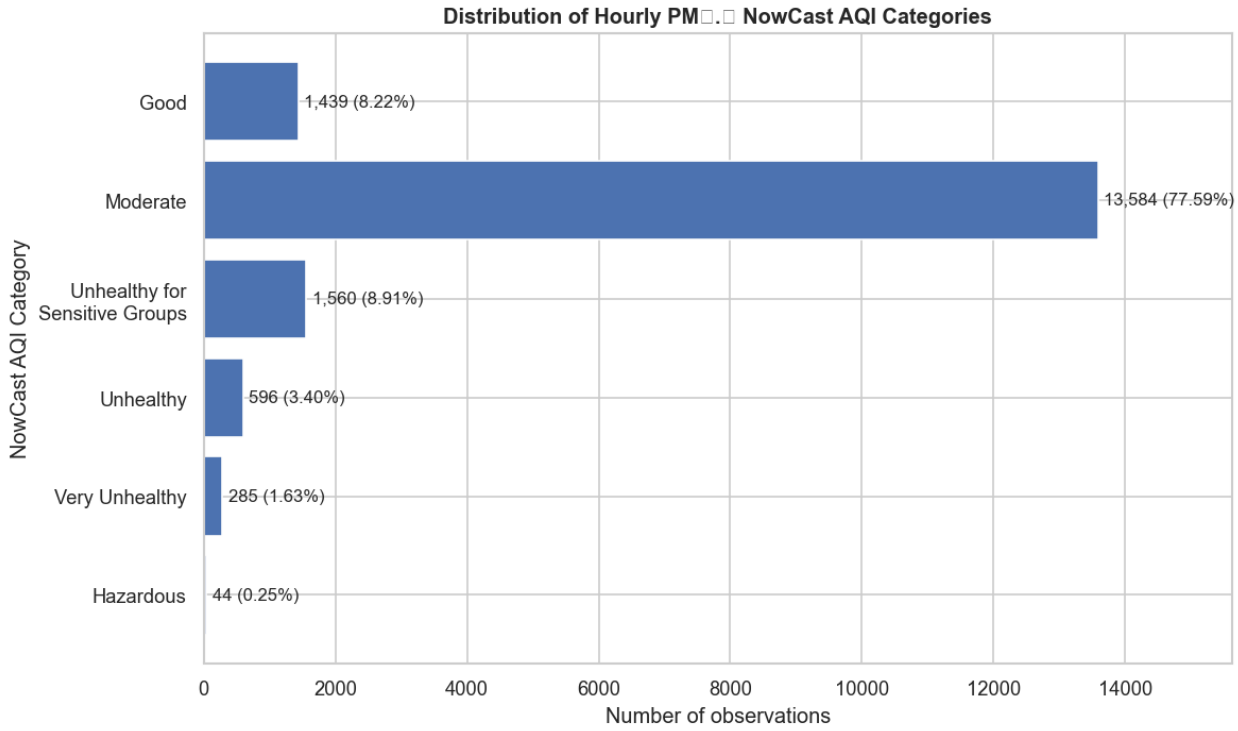


Fig.4. 3. Distribution of NowCast AQI categories.

Among the evaluated configurations, the RFE plus controlled-SMOTE model achieved the highest mean Accuracy (92.10%) and Weighted F1-score (90.86%). However, the RFE plus class-weighted SVM produced superior Macro Recall (89.14%) and more stable Macro F1 performance across temporal folds. The pooled Macro F1 and Macro Recall achieved by this configuration were 77.34% and 91.43%, respectively.

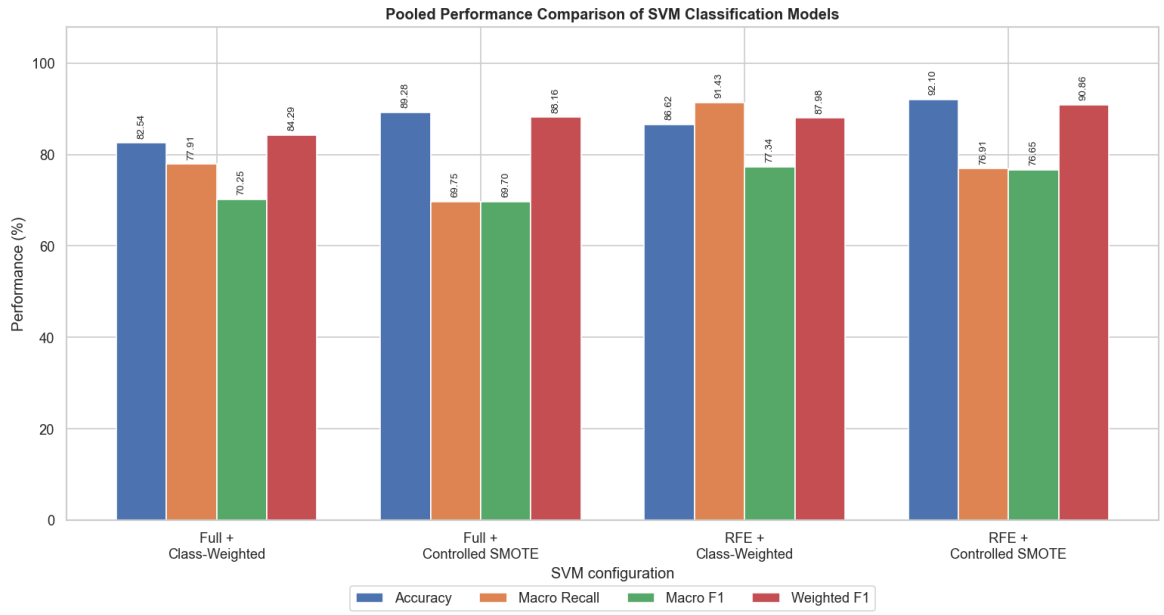


Fig.4. 4. Pooled performance comparison of SVM classification models

As shown in Figure 4, the results demonstrate a trade-off between maximizing overall accuracy and improving minority-class detection. Because balanced identification of higher-risk AQI categories was prioritized, the class-weighted SVM configuration was selected as the preferred classifier.

4.5 Annual Exposure and Health-Burden Estimates

The selected Random Forest model generated 8,760 hourly $PM_{2.5}$ forecasts for 2025. The predicted annual mean concentration was 20.57 µg/m³, closely matching the observed annual mean of 20.47 µg/m³ and yielding a relative exposure error of 0.48%.

Estimated relative risks ranged from 1.13 for COPD to 1.24 for ischemic heart disease (IHD) and lower respiratory infection (LRI). Corresponding population attributable fractions ranged between 11.29% and 19.56%.

Table 4. 3. Estimated $PM_{2.5}$-attributable health burden for Ghana in 2025 based on predicted annual exposure.

Outcome	Relative Rsik (RR)	PAF (%)	Attributable Deaths	Attributable DALYs
IHD	1.2431	19.5553	3,198.28	87,106.52
LRI	1.2431	19.5553	3,478.43	141,393.02
Stroke	1.1764	14.9968	4,311.54	132,787.29
COPD	1.1273	11.2923	241.46	9,052.06

Across the four disease outcomes, long-term $PM_{2.5}$ exposure was associated with an estimated burden of approximately **11,230 attributable deaths** and **370,339 DALYs**. Stroke contributed the

largest estimated mortality burden, whereas lower respiratory infection accounted for the largest DALY burden.

4.6 Discussion

4.6.1 Temporal Variability and Air-Quality Conditions

The monitoring record revealed substantial temporal variability in $PM_{2.5}$ concentrations, with pollution levels significantly elevated during the November-February period. The concentration peaks observed during these months are consistent with the influence of Harmattan conditions, which are characterized by increased atmospheric dust loading, reduced visibility, and enhanced pollutant accumulation. The concentration of exceedance events within the dry season indicates that seasonal atmospheric processes play a dominant role in shaping local air-quality conditions.

The finding that 14.09% of daily observations exceeded Ghana's 24-hour $PM_{2.5}$ standard highlights the persistence of episodic pollution events at the monitoring location. While annual averages provide an indication of long-term exposure, the observed short-term peaks demonstrate the importance of high-resolution monitoring for identifying periods of elevated health risk. These results further emphasize the need for forecasting systems capable of providing early warning of adverse air-quality conditions.

4.6.2 Forecasting Performance and Model Behaviour

Tree-based ensemble models consistently outperformed the evaluated deep-learning architectures for one-hour-ahead $PM_{2.5}$ forecasting. Random Forest achieved the strongest overall performance, followed closely by XGBoost, while the hybrid XGBoost-LSTM model performed better than the standalone CNN and LSTM architectures but did not exceed the tree-based approaches.

The superior performance of Random Forest and XGBoost likely reflects the structure of the predictor set used in this study. Extensive feature engineering incorporated lagged $PM_{2.5}$ observations, rolling statistical summaries, temporal indicators, meteorological variables, and interaction terms, thereby explicitly encoding much of the temporal information required for short-term forecasting. Under these conditions, the additional sequence-learning capability of deep-learning architectures may provide limited incremental benefit.

These findings do not imply that tree-based approaches universally outperform deep-learning models. Rather, they demonstrate that model performance remains highly dependent on data characteristics, predictor representation, forecasting horizon, and validation strategy. For environmental datasets with well-engineered predictors and moderate sample sizes, ensemble tree-based methods may provide a favourable balance between predictive accuracy, computational efficiency, and interpretability.

Feature-importance analysis further supports this interpretation. PM_{10}, lagged $PM_{2.5}$ variables, rolling statistical features, and meteorological interactions emerged as influential predictors, indicating that both pollution persistence and atmospheric conditions contribute substantially to

short-term $PM_{2.5}$ dynamics. However, these relationships should be interpreted as predictive associations rather than evidence of pollutant sources or causal mechanisms.

4.6.3 AQI Classification and Risk Communication

The AQI classification framework complements continuous pollutant forecasting by translating quantitative concentration estimates into categories that are more readily interpretable by policymakers and the public. The dominance of the Moderate AQI category, together with the limited representation of Very Unhealthy and Hazardous conditions, created a highly imbalanced classification problem.

The comparison between controlled-SMOTE and class-weighted SVM approaches revealed an important trade-off between maximizing overall accuracy and improving minority-class detection. Although the controlled-SMOTE configuration achieved the highest overall accuracy, the class-weighted approach produced superior Macro Recall and Macro F1-score, indicating a stronger ability to identify relatively rare but potentially important air-quality conditions. Because environmental-health applications often prioritize the detection of high-risk events, balanced performance across classes was considered more important than overall accuracy alone.

These findings demonstrate the importance of using evaluation metrics that account for class imbalance. Reliance solely on accuracy could overestimate model effectiveness in situations where the majority class dominates observations.

4.6.4 Implications for Environmental Health Assessment

A key contribution of this study is the integration of pollutant forecasting, AQI classification, and health-burden estimation within a single analytical framework. Most air-quality studies focus on predictive performance alone, whereas the present framework extends forecasting outputs into public-health assessment.

The close agreement between predicted and observed annual $PM_{2.5}$ concentrations suggests that the selected Random Forest model provided a reasonable exposure estimate for downstream epidemiological analysis. Using established concentration-response relationships and Ghana-specific burden estimates, long-term $PM_{2.5}$ exposure was associated with substantial attributable mortality and DALY burdens across cardiovascular and respiratory diseases. Stroke contributed the largest estimated mortality burden, while lower respiratory infection contributed the largest DALY burden.

These estimates should be interpreted as model-based population-level assessments rather than direct measures of disease occurrence. The calculated burden depends on assumptions regarding concentration-response coefficients, counterfactual exposure levels, baseline disease rates, and the representativeness of the monitoring location. Nevertheless, the results illustrate how air-quality forecasting can be translated into information relevant for environmental-health planning and policy evaluation.

More broadly, the framework demonstrates the potential value of combining machine-learning methods with epidemiological assessment to support air-quality management in resource-constrained environments. By linking environmental monitoring to public-health outcomes, such approaches can improve risk communication, strengthen evidence-based decision making, and support the development of early-warning systems for pollution-related health risks.

5. Limitations

This study has several limitations. First, the analysis was based on a single monitoring station and may not fully represent spatial variations in $PM_{2.5}$ exposure across the Greater Accra Metropolitan Area or Ghana. Second, the health-burden assessment combined site-specific exposure estimates with national-level GBD mortality and DALY statistics rather than spatially matched health records. Third, the estimates relied on published concentration-response relationships and a fixed theoretical minimum risk exposure level (TMREL). Finally, uncertainty associated with exposure prediction, relative-risk assumptions, and baseline disease burdens was not explicitly propagated through the epidemiological analysis. Therefore, the reported mortality and DALY values should be interpreted as indicative population-level estimates rather than precise measures of attributable disease burden.

Despite these limitations, the study demonstrates the feasibility of integrating $PM_{2.5}$ forecasting, AQI classification, and health-burden assessment within a unified environmental-health framework for resource-constrained settings.

6. Conclusions

This study developed an integrated environmental-health framework that combined high-resolution air-quality monitoring, one-hour-ahead $PM_{2.5}$ forecasting, AQI classification, and epidemiological health-burden estimation. The results demonstrated substantial temporal and seasonal variability in $PM_{2.5}$ concentrations, with the highest pollution levels occurring during the November-February period. These findings highlight the importance of seasonal influences on air quality and population exposure.

Among the evaluated forecasting models, the optimized Random Forest achieved the best overall predictive performance, while the hybrid XGBoost-LSTM was the strongest-performing deep-learning architecture. The results further showed that model performance benefited from the inclusion of a larger and more diverse predictor set, highlighting the importance of feature engineering in short-term $PM_{2.5}$ forecasting. For AQI classification, the class-weighted SVM provided the most balanced performance across imbalanced air-quality categories, achieving strong Macro Recall and stable classification performance.

The selected Random Forest model preserved continuous exposure information and enabled estimation of $PM_{2.5}$-attributable mortality and DALYs using Ghana-specific GBD 2023 baseline health data. By linking pollutant forecasting, air-quality risk communication, and health-impact assessment within a single analytical pipeline, the framework extends beyond conventional forecasting studies and demonstrates how environmental data can be translated into meaningful public-health information.

Overall, the study provides a practical and reproducible approach for supporting air-quality management, risk communication, and environmental-health decision-making in resource-constrained settings. Future research should incorporate spatially distributed monitoring networks, uncertainty quantification, and locally resolved health data to further improve the robustness and generalizability of the framework.

Statement and Declaration

Data Availability

The environmental data were obtained through the Afri-SET monitoring programme at the University of Ghana. Health-burden baseline estimates were derived from the IHME Global Burden of Disease 2023 database. Raw environmental and health data remain subject to the access conditions and terms of the respective data custodians. The analyses reported in this study were conducted using the derived datasets described in the Methods section.

Ethics Statement

This study used environmental monitoring data and aggregated, population-level health estimates. No individual-level patient identifiers, clinical records, or individually identifiable health information were analysed. Therefore, the study did not involve direct research participation by human subjects.

AI/LLM disclosure

Use of AI-assisted tools: During manuscript preparation, generative AI tools were used for language refinement, grammatical editing, and/or organizational assistance. The author reviewed and verified the content and remains fully responsible for the accuracy, interpretation, and integrity of the manuscript.

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